

$S \rightarrow Lb$

$$B \rightarrow \neg B \neg$$

B → D 2/

$B \rightarrow S$

$$\Sigma = \{a, b, c, \emptyset\}$$
$$V = \{s, L, B\}$$

$S \rightarrow Lb$

$$B \rightarrow c \quad B - c$$

B → 9

$B \rightarrow S$

Handwritten diagram illustrating the construction of a sequence of sets S_0, S_1, S_2, S_3 and the final result $L(1)$.

The sets are defined as follows:

- $S_0 = \{a\}$
- $S_1 = \{a, c\}$
- $S_2 = \{a, c\}$
- $S_3 = \{a, c\}$

The diagram shows the progression of the sequence and the final result $L(1) = \{a, c\}$.

$$No \quad L(1)$$

	0	1	2	3
S	$\{\checkmark\}$	$\{\checkmark, c\}$	$\{\checkmark, c\}$	$\{\checkmark, c\}$
L	$\{b\}$	$\{\checkmark, b\}$	$\{\checkmark, b, c\}$	$\{\checkmark, b, c\}$
B	$\{c\}$	$\{c\}$	$\{c\}$	$\{c\}$

$S \rightarrow aL$

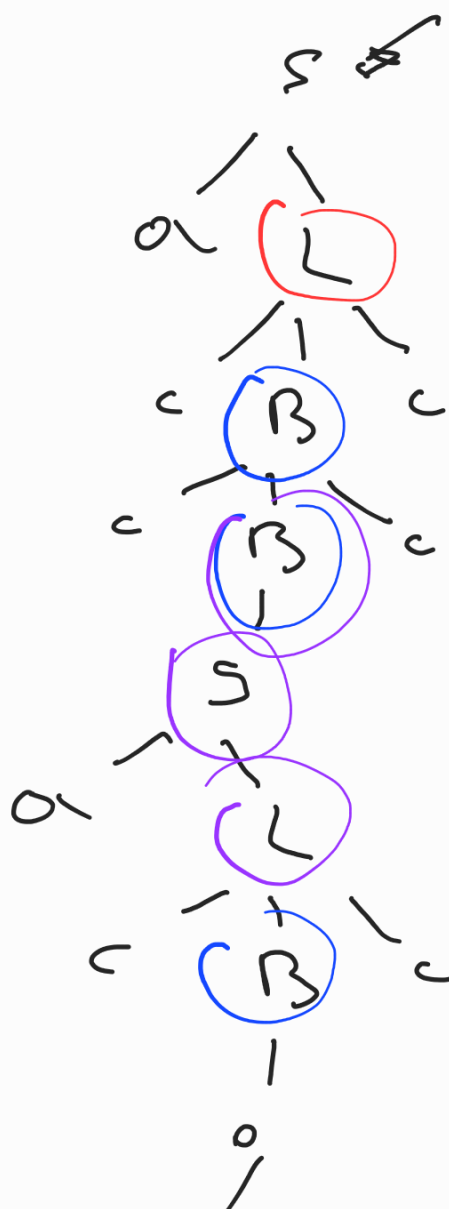
$S \rightarrow Lb$

$L \rightarrow -cB-c$

$B \rightarrow -cB-c$

$B \rightarrow \epsilon$

$B \rightarrow S$



$S \rightarrow L$
 $L \rightarrow a$
 $L \rightarrow D$
 $D \rightarrow aN$
 $D \rightarrow bN$
 $N \rightarrow \epsilon$
 $N \rightarrow m$
 $N \rightarrow z$
 $z \rightarrow mL$

$\Sigma = \{a, b, m\}$
 $V = \{S, L, D, N, z\}$

	0	1	2	3	Grm
$S \rightarrow L$	$\emptyset$	$\{a\}$	$\{a, b\}$	$\{a, b\}$	$\{a, b\}$
$L \rightarrow a$	$\{a\}$	$\{a\}$	$\{a\}$	$\{a\}$	$\{a\}$
$L \rightarrow D$	$\emptyset$	$\{a, b\}$	$\{a, b\}$	$\{a, b\}$	$\{a, b\}$
$D \rightarrow aN$	$\{a\}$	$\{a\}$	$\{a\}$	$\{a, b\}$	$\{a, b\}$
$D \rightarrow bN$	$\{b\}$	$\{b\}$	$\{b\}$	$\{a, b\}$	$\{a, b\}$
$N \rightarrow \epsilon$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\{a\}$
$N \rightarrow m$	$\{m\}$	$\{m\}$	$\{m\}$	$\{m\}$	$\{a\}$
$N \rightarrow z$	$\emptyset$	$\{m\}$	$\{m\}$	$\{m\}$	$\{a\}$
$z \rightarrow mL$	$\{m\}$	$\{m\}$	$\{m\}$	$\{m\}$	$\{a, b\}$

	0	1	2	3	4	5
S	$\{a\}$	$\{a\}$	$\{a\}$	$\{a\}$	$\{a\}$	$\{a\}$
L	$\emptyset$	$\{a\}$	$\{a\}$	$\{a\}$	$\{a\}$	$\{a\}$
D	$\emptyset$	$\emptyset$	$\{a\}$	$\{a\}$	$\{a\}$	$\{a\}$
N	$\emptyset$	$\emptyset$	$\emptyset$	$\{a\}$	$\{a\}$	$\{a\}$
z	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\{a\}$	$\{a\}$

$N \rightarrow n$	
$N \rightarrow z$	
$z \rightarrow n \quad L$	
$n \in L$	

$N \rightarrow n \quad X$	
$X \rightarrow \varepsilon$	$\{ \varepsilon \}$
$X \rightarrow L$	$\{ a, b \}$
$n \in L$	

XML

σ : $\langle \text{Name} \rangle$	a : attrName
m : $\rangle$	s : string '...' & '"'
e : $\langle / \rangle$	u : $=$
$-c$: $\langle / \text{Name} \rangle$	t : free text

$S \rightarrow E$

$E \rightarrow \sigma L e$

$E \rightarrow \sigma L m C e$

$L \rightarrow E$

$L \rightarrow a u s L$

$C \rightarrow E$

$C \rightarrow t C$

$C \rightarrow E C$

$S \rightarrow E$

$E \rightarrow \sigma L a$

$E \rightarrow \sigma L m C \tau$

$L \rightarrow E$

$L \rightarrow a m \Delta L$

$C \rightarrow E$

$C \rightarrow t C$

$C \rightarrow E C$

	0	1	2	Grm
S	$\emptyset$	$\{\sigma\}$	$\{\sigma\}$	$\{\sigma\}$
E	$\{\sigma\}$	$\{\sigma\}$	$\{\sigma\}$	$\{\sigma\}$
E	$\{\sigma\}$	$\{\sigma\}$	$\{\sigma\}$	$\{\sigma\}$
L	$\emptyset$	$\emptyset$	$\emptyset$	$\{\sigma\}$
L	$\{a\}$	$\{a\}$	$\{a\}$	$\{a, m\}$
C	$\emptyset$	$\emptyset$	$\emptyset$	$\{a\}$
C	$\{\tau\}$	$\{\tau\}$	$\{\tau\}$	$\{\tau\}$
C	$\emptyset$	$\{\sigma\}$	$\{\sigma\}$	$\{\tau\}$
				$\{\sigma\}$

	0	1	2
S	$\{\sigma\}$	$\{\sigma\}$	$\{\sigma\}$
E	$\{\sigma, \tau\}$	$\{\tau, \sigma, t, \sigma\}$	$\{\tau, \sigma, t, \sigma\}$
L	$\{a, m\}$	$\{a, m\}$	$\{a, m\}$
C	$\{\tau\}$	$\{\tau\}$	$\{\tau\}$

NO $L(1)$

$$S \rightarrow E$$

$$E \rightarrow \sigma L X$$

$$X \rightarrow e$$

$$X \rightarrow n C c$$

$$\underline{\{ \sigma \}}$$

$$\{ e \}$$

$$\underline{\{ n \}}$$

$$L \rightarrow E$$

$$L \rightarrow a n s L$$

$$C \rightarrow E$$

$$C \rightarrow t C$$

$$C \rightarrow E C$$

$$E' \quad L L (1)$$